

A smooth one-dimensional counterexample to coordinate independence in a global implicit-function growth bound

Candidate counterexample packet for arXiv:2004.04427

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Status. Candidate full counterexample, likely valid, pending human review. The growth condition in Theorem 2.1(iii) of the source is not independent of the chosen diffeomorphisms, even for the equation $x - y = 0$ on $\mathbb{R}^2$ and smooth coordinate changes of $\mathbb{R}$.

1 Source question

The source is T. Berger and F. Haller, *On an extension of a global implicit function theorem*, arXiv:2004.04427, published in *Comptes Rendus Mathématique* 360 (2022), 439–450 [1]. Remark 2.3 on source PDF page 7 asks whether the growth bound in Theorem 2.1(iii) is independent of the diffeomorphisms ϕ and ψ . More precisely, if the bound holds for one pair of coordinates, must it hold for any other diffeomorphisms $\hat{\phi}, \hat{\psi}$, with some continuous weight $\hat{\omega}$? The remark explicitly leaves open whether this is true or whether a counterexample exists.

2 The condition to be tested

In the one-dimensional setting, a *continuous weight* is a continuous nondecreasing function $\omega : [0, \infty) \rightarrow (0, \infty)$ satisfying the Osgood condition

$$\int_0^\infty \frac{dt}{\omega(t)} = \infty. \quad (1)$$

For $F : X \times Y \rightarrow \mathbb{R}$, a zero set Z , a left inverse S of $D_y F$, and diffeomorphisms ϕ, ψ of the two projections, the source growth bound is

$$|D\psi(y)S(x, y)| |D_x F(x, y)(D\phi(x))^{-1}| \leq \omega(|\psi(y)|), \quad (x, y) \in Z. \quad (2)$$

3 Proof intuition

A global increasing diffeomorphism $h : \mathbb{R} \rightarrow \mathbb{R}$ cannot have a derivative that is uniformly too large as a function of its output: writing $t = h(y)$, the inverse derivative must integrate to infinite length. This does not, however, prevent h' from having extremely high, extremely narrow spikes.

We prescribe the output-coordinate speed $H(t) = h'(h^{-1}(t))$ to equal 1 except on intervals of summable total length, while making $H(n) \geq 4^n$ at every positive integer. The large complement on which $H = 1$ guarantees that $\int_0^\infty H^{-1} = \infty$, so the resulting h is a global smooth diffeomorphism.

Remark 2.3. An important question that arises is whether the growth bound in condition (iii) in Theorem 2.1 is independent of the choice of the diffeomorphisms ϕ and ψ . Use the notation from Theorem 2.1, assume that conditions (i)–(iii) are satisfied and let $\hat{\phi} : \pi_1(Z) \rightarrow \hat{\mathcal{X}}$ and $\hat{\psi} : \pi_2(Z) \rightarrow \hat{\mathcal{Y}}$ be diffeomorphisms for some Banach spaces $\hat{\mathcal{X}}, \hat{\mathcal{Y}}$. Then, omitting the subscripts indicating the spaces corresponding to the norms, we have the estimate

$$\begin{aligned} & \left\| D\hat{\psi}(y) \cdot S(x, y) \right\| \cdot \left\| D_x F(x, y) \cdot (D\hat{\phi}(x))^{-1} \right\| \\ & \leq \left\| D\psi(y) \cdot S(x, y) \right\| \cdot \left\| D\hat{\psi}(y) \cdot (D\psi(y))^{-1} \right\| \\ & \quad \cdot \left\| D_x F(x, y) \cdot (D\phi(x))^{-1} \right\| \cdot \left\| D\phi(x) \cdot (D\hat{\phi}(x))^{-1} \right\| \\ & \leq \omega(\|\psi(y)\|_{\mathcal{Y}}) \cdot \left\| D\hat{\psi}(y) \cdot (D\psi(y))^{-1} \right\| \cdot \left\| D\phi(x) \cdot (D\hat{\phi}(x))^{-1} \right\| \end{aligned}$$

for all $(x, y) \in Z$. If the last term satisfies

$$\begin{aligned} \forall (x, y) \in Z : \omega(\|\psi(y)\|_{\mathcal{Y}}) \cdot \left\| D\hat{\psi}(y) \cdot (D\psi(y))^{-1} \right\| \cdot \left\| D\phi(x) \cdot (D\hat{\phi}(x))^{-1} \right\| \\ \leq \hat{\omega}(\|\hat{\psi}(y)\|_{\hat{\mathcal{Y}}}) \end{aligned}$$

for some continuous weight $\hat{\omega}$, then the growth bound in condition (iii) would indeed be independent of ϕ and ψ . However, it is still an open problem whether this is true (or a counterexample exists) and remains for future research.

Figure 1: Remark 2.3 on page 7 of the source PDF.

On the other hand, any nondecreasing function dominating H must stay above 4^n throughout $[n, n+1]$. Its reciprocal is therefore integrable, so it cannot be a weight in the sense of (1). Applied to $F(x, y) = x - y$, this changes an original growth bound identically equal to 1 into one that has no admissible weight majorant.

4 Counterexample

Theorem 1. *There exist data satisfying all hypotheses of Theorem 2.1 of the source and diffeomorphisms ϕ, ψ for which the growth condition (2) holds, together with alternative smooth diffeomorphisms $\hat{\phi}, \hat{\psi}$ for which it fails for every continuous weight. Consequently the answer to the question in Remark 2.3 is negative.*

Proof. Define the explicit nonnegative smooth bump

$$\beta(u) := \begin{cases} e \exp\left(-\frac{1}{1-16u^2}\right), & |u| < \frac{1}{4}, \\ 0, & |u| \geq \frac{1}{4}. \end{cases}$$

Thus $\beta \in C_c^\infty(\mathbb{R})$, $0 \leq \beta \leq 1$, and $\beta(0) = 1$. Set

$$H(t) := 1 + \sum_{n=1}^{\infty} 4^n \beta(4^n(t-n)). \quad (3)$$

The sum is locally finite, hence H is smooth and $H \geq 1$. The n th bump is supported in

$$I_n := (n - 4^{-(n+1)}, n + 4^{-(n+1)}).$$

The union $U := \bigcup_{n \geq 1} I_n$ has finite measure, since

$$|U| \leq \sum_{n=1}^{\infty} 2 \cdot 4^{-(n+1)} = \frac{1}{6}.$$

Outside U one has $H = 1$. It follows that

$$\int_0^\infty \frac{dt}{H(t)} = \infty, \quad (4)$$

because $[0, \infty) \setminus U$ has infinite measure. On the negative half-line $H = 1$ as well.

Now define

$$Q(t) := \int_0^t \frac{ds}{H(s)}.$$

The function Q is smooth and strictly increasing, with $Q'(t) = H(t)^{-1} > 0$. It tends to $-\infty$ as $t \rightarrow -\infty$ and, by (4), to $+\infty$ as $t \rightarrow +\infty$. Hence it is a smooth diffeomorphism of $\mathbb{R}$ onto $\mathbb{R}$. Let

$$h := Q^{-1}.$$

The inverse-function formula gives

$$h'(y) = H(h(y)). \quad (5)$$

We next supply the implicit-function data. Take

$$X = Y = \mathbb{R}, \quad F(x, y) = x - y, \quad Z = F^{-1}(0) = \{(x, x) : x \in \mathbb{R}\}.$$

Here the ambient Banach spaces are all $\mathbb{R}$. The set Z is closed and path-connected. Since $D_y F = -1$, the constant operator $S = -1$ is a left inverse: $S D_y F = 1$. Choose the original diffeomorphisms $\phi = \psi = \text{id}_{\mathbb{R}}$. Then the left side of (2) is identically

$$|1 \cdot (-1)| |1 \cdot 1| = 1.$$

Thus condition (iii) holds with the constant weight $\omega \equiv 1$, which satisfies (1). All hypotheses of the source theorem are therefore satisfied.

Finally choose the alternative diffeomorphisms

$$\widehat{\phi} = \text{id}_{\mathbb{R}}, \quad \widehat{\psi} = h.$$

At $(x, y) \in Z$, the new left side of the growth bound is, by (5),

$$|h'(y)(-1)| |1| = H(h(y)). \quad (6)$$

Suppose that some continuous weight $\widehat{\omega}$ made the new bound valid. Since h maps $\mathbb{R}$ onto $\mathbb{R}$, putting $t = h(y)$ in (6) would give

$$H(t) \leq \widehat{\omega}(|t|) \quad (t \in \mathbb{R}). \quad (7)$$

At every positive integer, (3) and $\beta(0) = 1$ give $H(n) \geq 4^n$. Since $\widehat{\omega}$ is nondecreasing, (7) implies

$$\widehat{\omega}(t) \geq \widehat{\omega}(n) \geq H(n) \geq 4^n, \quad n \leq t < n + 1.$$

Consequently

$$\int_0^\infty \frac{dt}{\widehat{\omega}(t)} \leq \int_0^1 \frac{dt}{\widehat{\omega}(t)} + \sum_{n=1}^\infty 4^{-n} < \infty.$$

This contradicts the defining condition (1). No admissible weight exists for the alternative coordinates, completing the counterexample. $\square$

5 Verification report

The construction is analytic and uses no numerical premise. The following points were checked explicitly:

- The supports I_n are disjoint and locally finite; hence the series in (3) defines a smooth positive function.
- Their total length is $1/6$, so the set on which $H = 1$ has infinite measure. This proves the global surjectivity of Q , rather than merely a local inverse statement.
- The identity (5) follows directly from differentiating $Q(h(y)) = y$.
- The original data satisfy each item of Theorem 2.1, and the original constant weight is admissible.
- The contradiction uses the required monotonicity of a weight. A non-monotone function could follow the narrow spikes, but such a function is not a weight in the source’s definition.

The main human-review focus should be the quantifiers in Remark 2.3: the original pair of diffeomorphisms satisfies condition (iii), whereas the explicit alternative pair admits no weight at all. Current verdict: **full counterexample likely valid**.

6 Scope and novelty check

This is a counterexample to coordinate-independence of the sufficient growth condition, not to Theorem 2.1 itself. Indeed the underlying equation has the global implicit function $y = x$ in every coordinate system. The example shows that the theorem’s hypothesis can be destroyed by a smooth reparametrization even though its conclusion remains true.

Local indexes were searched for arXiv:2004.04427, the title, “global implicit function”, “growth bound”, and “diffeomorphism independence”, with no prior result. A bounded web/arXiv search on 21 July 2026 used the title, authors, DOI, exact Remark 2.3 wording, and close “continuous weight”, “coordinate independence”, and “counterexample” combinations. It found the source and citations but no settlement; the March 2026 arXiv text still calls the question open. This search was bounded, not exhaustive.

References

- [1] T. Berger and F. Haller, *On an extension of a global implicit function theorem*, Comptes Rendus Mathématique 360 (2022), 439–450; arXiv:2004.04427. DOI: 10.5802/crmath.309.